SELECTION

Background Information

It is a generally accepted principle in contemporary psychology that one of, if not the, best predictors of future behavior is the individual's past behavior, especially in similar settings. It is not surprising, therefore, that background data on prospective employees have in fact proven to be consistently superior to that obtained from the two other most widely used selection methods, interviewing and testing.

The usefulness of personal history data as a predictor of future behavior was underscored early in psychology, particularly by Allport's (1942) emphasis on the usefulness for such purposes of personal documents (letters, diaries, autobiographies) and Guthrie's (1944) stress on the predictive value of past affiliations. In American industry around the same period there began early efforts to objectively score employee application blanks. In a number of investigations, especially a long-term series of studies conducted at Standard Oil (Henry, 1966), such application-blank information as school achievement, leadership, early family relationships, personal goals, health, and leisure-time activities indeed proved predictively valuable in forecasting work performance. These several demonstrations led many companies to more fully objectify their application blanks, giving rise to the scored biodata form. Such forms, essentially standardized and objectively scorable, background information, application questionnaires, have found widespread acceptance in American industry. The Biographical Information Blank, the Biographical Data Form, and the Individual Background Survey are a few current examples. Cascio (1982) nicely summarized the important role of such information in the selection process.

> What is impressive is that indicators of past successes and accomplishments can be utilized in an objective way to identify persons with differing odds of being successful over the long term in their management career. People who have a prior record of achievement when they enter an organization are in excellent positions to profit from training opportunities and from challenging organizational environments. (p. 196)

The Interview

Although very serious questions have been raised about the validity of information gathered in the employment interview for predicting future on-the-job performance, this nevertheless continues to be the single most widely used selection technique. In fact, its use has even increased in recent years (in

the face of an array of findings of bias, weakness, and poor validity about the interview), probably because of the difficult legal challenges raised in the last decade to a companion selection technique, psychological testing.

What are these weaknesses and sources of predictive invalidity? Several investigators (Carlson, 1970; Hakel, Ohnesorge, & Dunnette, 1970; Landy, 1973) have shown how powerful are the initial biases about the interviewee that the interviewer can bring to the interview, perhaps originating from an unfavorable letter of recommendation or an application-blank statement. Under such circumstances, the interviewee could be given less credit for past accomplishments, be held more personally responsible for past failures, and in the end be less likely to be offered the job for which she or he is applying.

A second preinterview source of interviewer bias is the other applicants interviewed immediately before the present one. Strong contrast effects often operate here. The "just average" job candidate interviewed immediately following two or three clearly below average ones tends to be evaluated very positively by the interviewer. Contrastingly, being interviewed following outstanding applicants often makes the applicant appear weaker than his or her own qualifications warrant. Once the interview actually begins, the first few minutes play a crucial role in influencing the interviewer's perceptions, observations, and later decisions about acceptance. Important here in determining the favorableness or unfavorableness of this *primacy effect* is the applicant's appearance and interpersonal skills when greeting the interviewer and in the opening discussion. The potential for interviewer bias continues throughout the interview. Hakel et al. (1970) noted that interviewers often devote considerable interview time to searching for negative information and give it disproportionate weight when they find it. Anderson (1960) found that interviewees were more likely to be hired after interviews in which the interviewer did a great deal of the talking. Though it is unclear whether this reflects interviewer bias, appropriate information exchange, or other factors, it is the case that the longer the interview, the more likely it is that the applicant will be hired. More generally, many preemployment interviews can be accurately described, as Cascio (1982) observed, as events in which interviewers develop a stereotype of a good applicant and then proceed to search for qualities that match or fail to match their stereotypes.

Given these several powerful effects of interviewer bias, as well as the fact that interviewers' behaviors and postinterview decisions can also be influenced by applicants' sex, educational level, age, physical attractiveness, and a host of nonverbal behaviors (Dipboye, Fromkin, & Wiback, 1975; Zikmund, Hitt, & Pickens, 1978), we must agree with Rowland and Ferris's (1982) depressing conclusion that:

> The research literature presents a veritable rogue's gallery of biases and other assorted effects through which interviewer decision-making can go wrong.

> Individually, these studies lead to pessimism about the possibility of re-engineering the interview. Collectively, their impact is almost devastating. (p. 140)

Attempts to diminish these several sources of interview invalidity by means of interviewer training have generally been unsuccessful. Wexley, Sanders, and Yukl (1973), for example, showed that neither the use of warnings (lectures) nor anchoring (comparing applicants to a pre-set standard rather than to each other) reduced the frequency of the applicant-to-applicant contrast effects just described. A more hopeful direction has been to seek to add structure and standardization to the interview process and, thus, in a sense to diminish somewhat the opportunity for interviewers' biases and related effects to operate. Evidence in recent years, therefore, has shown improved predictive validity for the preemployment interview when " . . . the content is focused on job-related situational questions, job knowledge questions, work requirement questions . . . " (Perloff, Craft, & Perloff, 1984, p. 429).

Before concluding our discussion of the selection interview, we wish to note that, in addition to its information-gathering purpose (for the interviewer), it can also intentionally and unintentionally serve an information-providing function (for the interviewee). Just as interviewers can be appropriately and inappropriately responsive to applicant characteristics, applicants can form too hasty and erroneous impressions of both the recruiting interviewer and the company he or she represents, as a result of interviewer age, sex, race, style of delivery, and job title (Rynes, Heneman, & Schwab, 1980). Particularly helpful to an applicant's forming favorable impressions is a recruiting interviewer who is well-prepared in her or his knowledge of both the applicant and the job being applied for and who, in addition, is or has been an actual incumbent in that job.

In spite of very considerable reservations, we would not wish to recommend that preemployment interviews be eliminated from the employee selection process. When used in a structured format, perhaps somewhat analogous to a verbally administered questionnaire, they can make an important and relatively unbiased contribution to employee-selection decision making. Their proper role beyond this limited one must await further research and refinement.

Psychological Testing

Of the many varied functions performed by the applied psychologist in industrial settings, those associated with psychological testing are clearly most frequent. Test development, test administration, test interpretation, and research on test validation are the primary examples. Although the first psychological tests were not created by psychologists (Gideon and Plato

advocated military aptitude testing centuries before psychology existed as a discipline), psychological testing has been with us for over 100 years.

In 1869, Galton began psychology's long interest in intelligence testing with his book *Hereditary Genius*, which presented a classification scheme based upon abilities. Shortly thereafter, Wundt began using reaction-time measures and tests of visual and auditory acuity as indexes of intelligence. This effort was greatly advanced by Binet in the same era, because he reconceptualized intelligence as consisting of more complex mental processes than had Wundt and, accordingly, developed tests of memory, attention, comprehension, imagination, and the like (see chapter 1). Munsterberg in 1913 brought this fledging movement into the industrial context in general, and employee selection in particular, by using a battery of tests available at the time to help select motormen for the Boston Railway Company. Psychological testing has blossomed fully since that early era, to gain a place of high prominence not only in industry, but also in school settings, clinical settings, and elsewhere. Its greatest boost for selection purposes came during World War I, at a time when the American Psychological Association (60,120 members in 1985) consisted of 336 people. Under the leadership of many of these early pioneers, 1,726,966 men were tested (for selection and classification purposes) as a part of the country's war effort.

In the period since these beginnings, the numbers and types of psychological tests employed in industry have expanded greatly. In a large, national survey done by the American Society for Personnel Administration in 1975, it was reported that approximately 60% of employers with over 25,000 employees and 40% of those with less than 100 employees used psychological tests. Current tests in major use follow.

Aptitude Tests

These tests measure the capacities of the individual for learning the skills and knowledge necessary to perform the job well.

Achievement Tests

Such tests measure proficiency or competencies already acquired in job skills and knowledge.

Interest Tests

Measured here are the activities of a vocational and avocational nature (e.g., mechanical, scientific, social welfare, clerical, business, esthetic) that an individual would prefer to engage in and enjoy engaging in.

Cognitive-Ability Tests

Descending from the early work on intelligence testing and the many efforts that followed to identify the factors or elements of which intelligence was

composed, these are measures of such abilities as word fluency, perceptual speed, memory, inductive reasoning, and comprehension.

Leadership Ability Tests

These tests have taken several forms, but most have sought to measure two central leadership dimensions: the task-oriented quality of initiating structure (defines and structures one's own and subordinates' roles) and the relationship-oriented quality of consideration (enhances mutual trust, shows respect for the ideas of others, considers others' feelings).

Personality Tests

These tests, as used in industry and elsewhere, have taken a particularly wide variety of forms. Designed to reflect enduring qualities, traits, or behavioral predispositions, their form has included questionnaires completed by others, self-report inventories, behavioral measures (described later), and projective techniques (i.e., ambiguous test stimuli onto which the person is held to cast or project individual qualities when responding).

Peer Assessment

These are test procedures by means of which selection and other information relevant to personnel decisions are obtained from other persons in, or applying to join, the work force. The three major forms of peer assessment are peer nomination (each person nominates one or more other persons as highest and lowest on a given dimension), peer rating (each member rates every other member on a given dimension or quality), and peer ranking (each member ranks every other member from best to worst).

Work-Sample Tests

Tests designed to obtain samples of the actual behaviors one is trying to predict have become quite widespread in contemporary psychology, and their major form in industrial settings has been work-sample tests, of which three in particular are especially popular. The *leaderless group discussion* is a procedure in which a group of candidates is asked to carry on a discussion for a period of time, no one is appointed leader, and raters observe and rate each person's performance. The *in-basket technique* is a procedure in which the testee is told about the organization involved, is asked to pretend he or she is now in a managerial position in that organization, and attempts to deal effectively with a variety of memos, letters, telephone calls, and other materials in an in-basket. *Business games* are a third prominent example of work-sample tests. These are primarily simulations in which the candidate is required to hire, fire, sell, deal with stress, or show other administrative skills in tasks consisting of human relations problems, running a warehouse, managing a stock market portfolio, and the like.